Lecture Notes: Chapter 3

Sections 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11

Spring, 2020

Contents

1

Section 3.1 - Introducing the Derivative

In section 2.1, we first looked at the notion of a tangent line. While we were only able to estimate the slope of the tangent line before, we now have the tools to define the tangent line precisely.

Definition 1 (Rate of Change and the Slope of the Tangent Line).
The average rate of change in f on the interval $[a,x]$ is the slope of the corresponding secant line:

$$m_{sec} = \frac{f(x)-f(a)}{x-a}.$$

The instantaneous rate of change in f at a is

$$m_{tan} = \lim_{x\to a}\frac{f(x)-f(a)}{x-a},$$

which is also the slope of the tangent line at $\big(a, f(a)\big)$, provided this limit exists. The tangent line is the unique line through $\big(a, f(a)\big)$ with slope m_{tan}. Its equation is

$$y - f(a) = m_{tan}(x-a)$$

Alternatively, we could define this as follows, which is frequently more useful in practice:

Definition 2 (Rate of Change and the Slope of the Tangent Line).
The average rate of change in f on the interval $[a,a+h]$ is the slope of the corresponding secant line:

$$m_{sec} = \frac{f(a+h)-f(a)}{h}.$$

The instantaneous rate of change in f at a is

$$m_{tan} = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h},$$

which is also the slope of the tangent line at $\big(a, f(a)\big)$, provided this limit exists.

The notion of the tangent line leads us to the concept of the derivative. Namely, if we can find the slope of the tangent line at $(x, f(x))$ for x in some open interval I, we can define a *function which gives us the slope of the tangent line* at each value of x in I. We make this precise as folows:

Definition 3 (The Derivative Function).
The derivative of f is the function

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h},$$

provided the limit exists and x is the domain of f. If $f'(x)$ exists, we say that f is differentiable at x. If f is differentiable at every point of an open interval I, we say f is differentiable on I.

Example 1. Let $f(x) = 7x - 3$. Find $f'(x)$, then find $f'(1)$, $f'(2)$, and $f'(3)$.

We use Definition 3 to find $f'(x)$:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(7(x+h) - 3) - (7x - 3)}{h} \\ &= \lim_{h\to 0} \frac{7x + 7h - 3 - 7x + 3}{h} \\ &= \lim_{h\to 0} \frac{7h}{h} \\ &= \lim_{h\to 0} 7 = 7. \end{aligned}$$

So, $f'(x) = 7$. Correspondingly, $f'(1) = f'(2) = f'(3) = 7$ as well.

Example 2. Let $f(x) = -3x^2 + 5x - 6$. Find $f'(x)$, then find $f'(1)$, $f'(2)$, and $f'(3)$.

We use Definition 3 to find $f'(x)$:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(-3(x+h)^2 + 5(x+h) - 6) - (-3x^2 + 5x - 6)}{h} \\ &= \lim_{h\to 0} \frac{-3x^2 - 6xh - 3h^2 + 5x + 5h - 6 + 3x^2 - 5x + 6}{h} \\ &= \lim_{h\to 0} \frac{-6xh - 3h^2 + 5h}{h} \\ &= \lim_{h\to 0} -6x - 3h + 5 = -6x + 5. \end{aligned}$$

So, $f'(x) = -6x + 5$. Correspondingly,

$$f'(1) = -6(1) + 5 = -1$$

$$f'(2) = -6(2) + 5 = -7$$

$$f'(3) = -6(3) + 5 = -13.$$

Example 3. Let $f(x) = \dfrac{2x}{x-5}$. Find $f'(x)$, then find $f'(1)$, $f'(2)$, and $f'(3)$.

We use Definition 3 to find $f'(x)$:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{2(x+h)}{(x+h)-5} - \frac{2x}{x-5}\right) \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{2x+2h}{x+h-5} - \frac{2x}{x-5}\right) \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{(x-5)(2x+2h)}{(x+h-5)(x-5)} - \frac{(x+h-5)(2x)}{(x+h-5)(x-5)}\right) \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{(x-5)(2x+2h)-(x+h-5)(2x)}{(x+h-5)(x-5)}\right) \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{(2x^2+2xh-10x-10h)-(2x^2+2xh-10x)}{(x+h-5)(x-5)}\right) \\
&= \lim_{h\to 0} \frac{1}{h}\left(\frac{-10h}{(x+h-5)(x-5)}\right) \\
&= \lim_{h\to 0} \frac{-10}{(x+h-5)(x-5)} = \frac{-10}{(x-5)^2}
\end{aligned}$$

So, $f'(x) = \dfrac{-10}{(x-5)^2}$. Correspondingly,

$$f'(1) = \frac{-10}{(1-5)^2} = -\frac{5}{8}$$

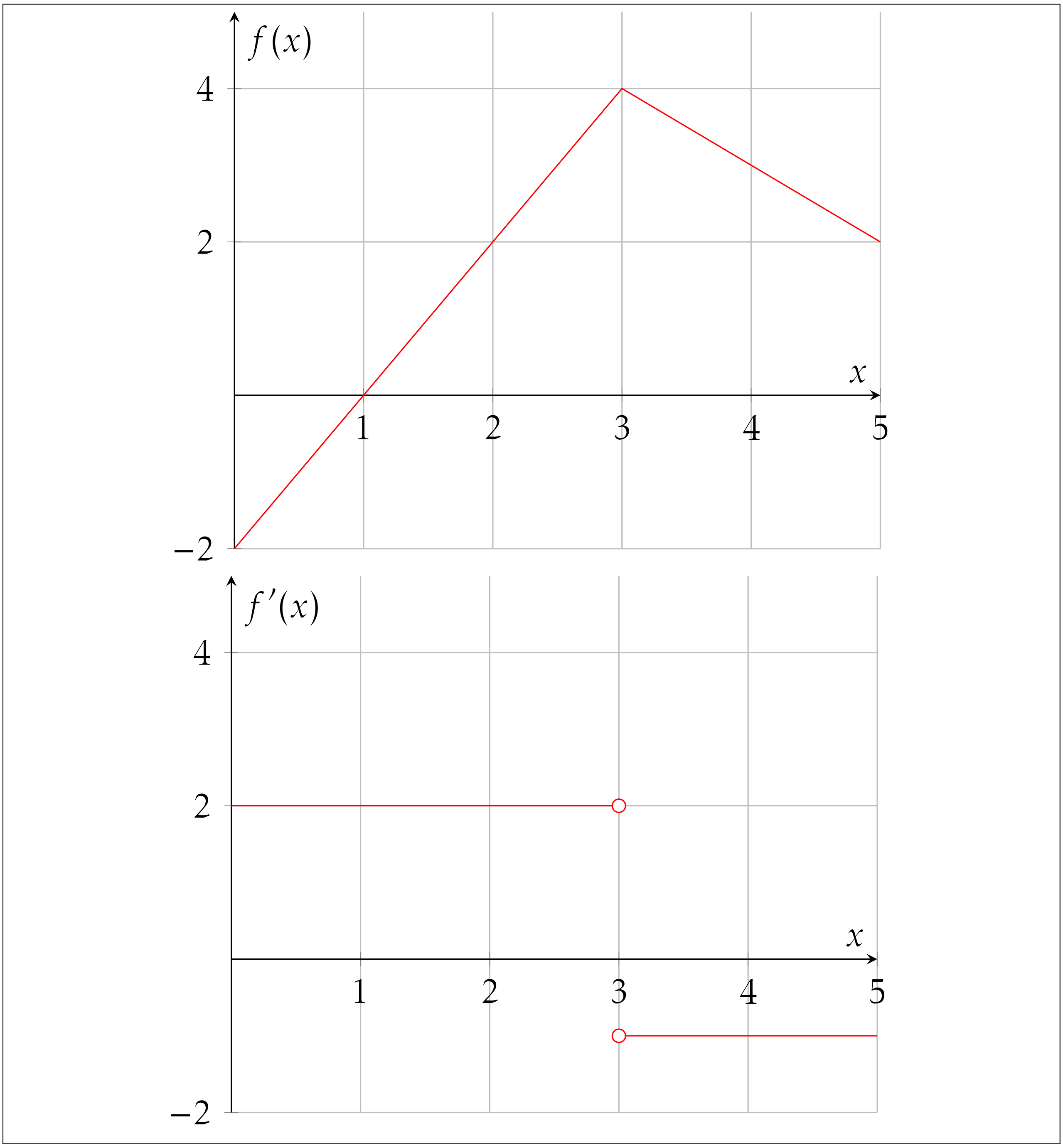
$$f'(2) = \frac{-10}{(2-5)^2} = -\frac{10}{9}$$

$$f'(3) = \frac{-10}{(3-5)^2} = -\frac{5}{2}.$$

Section 3.2 - Working With Derivatives

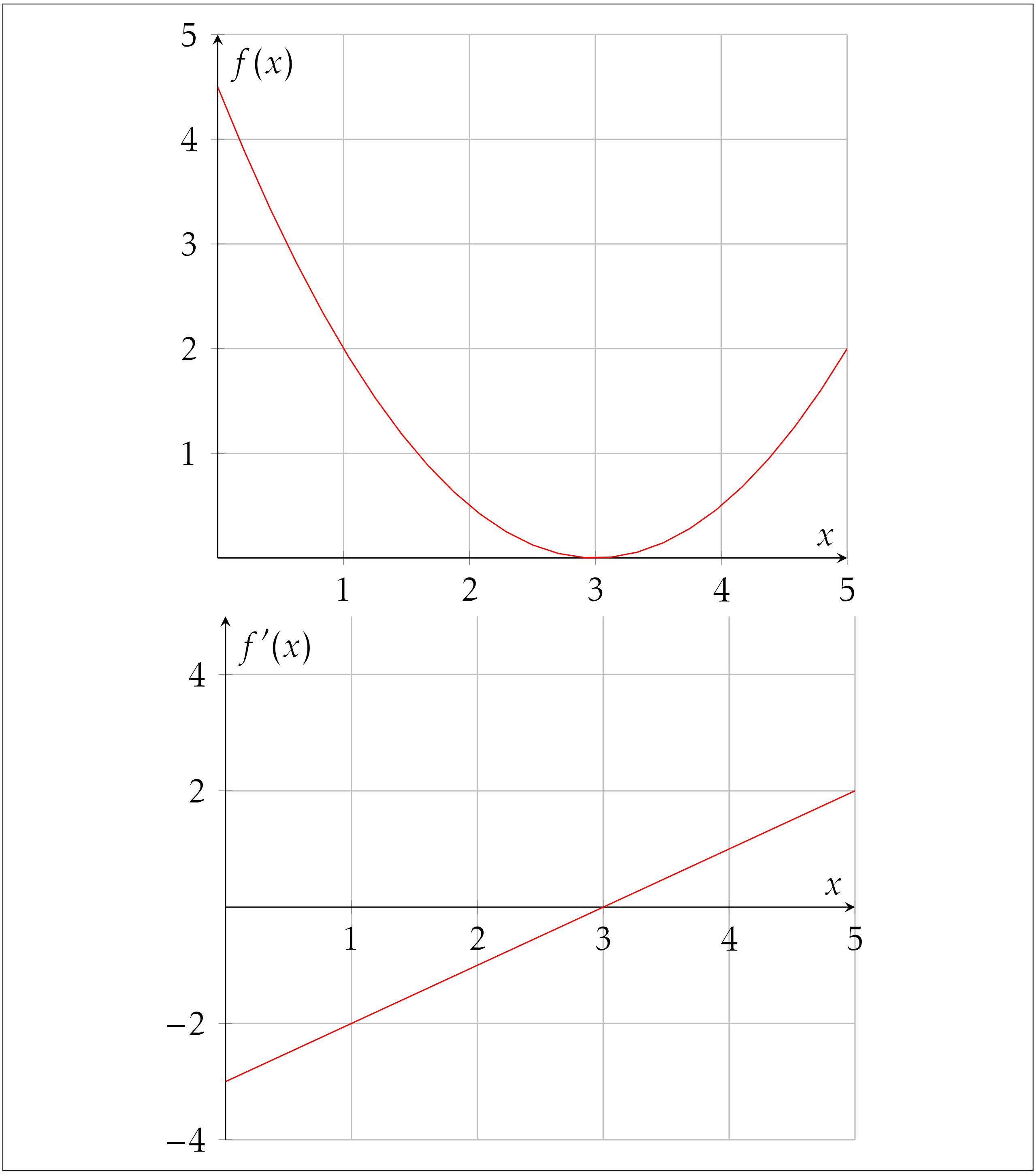
The function f' is called the derivative of f because it is derived from f. The following examples illustrate how to derive the graph of f' from the graph of f.

Example 4.
The graph of a function f is given below. Sketch a graph of f' using this information.



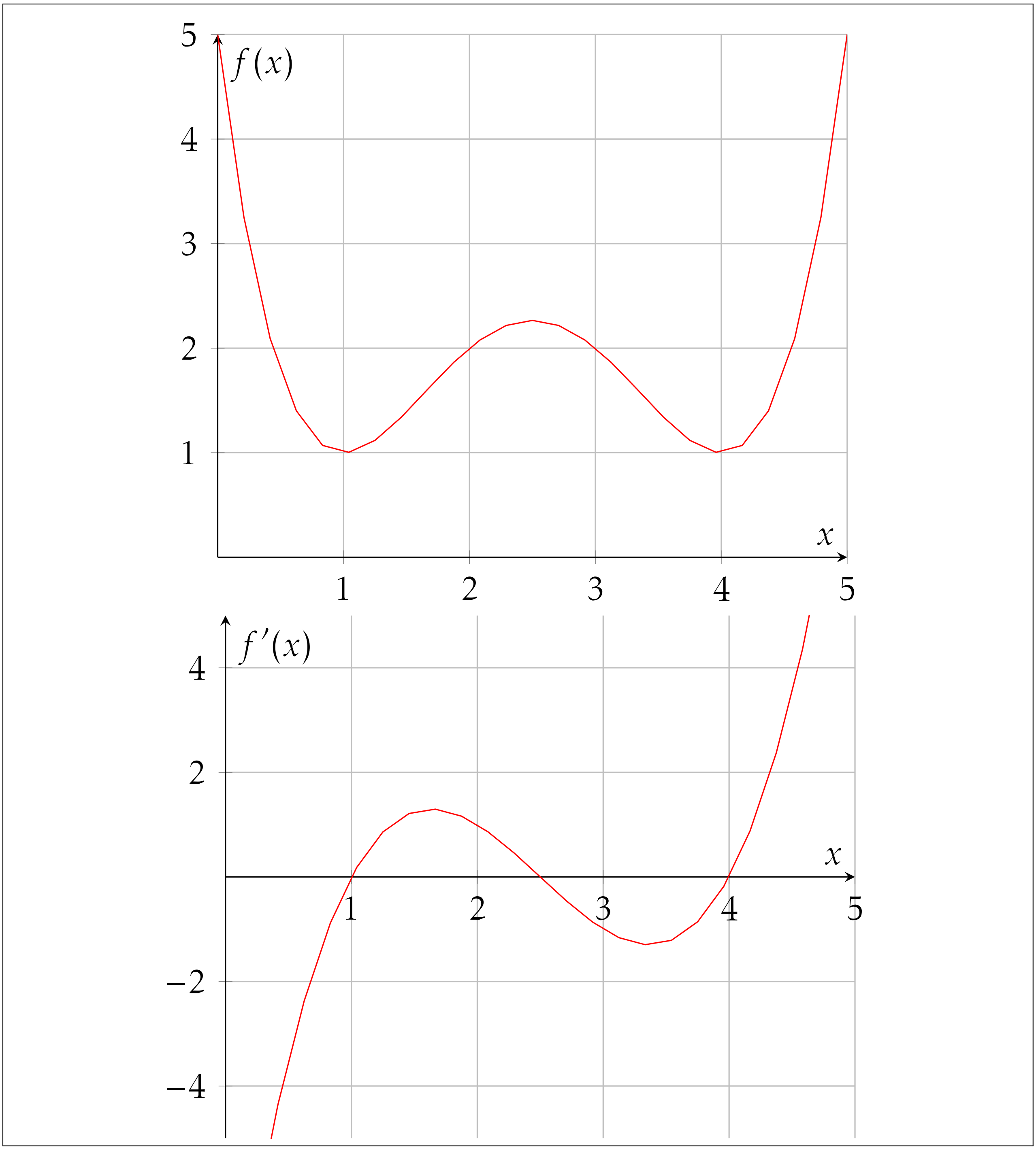
Example 5.

The graph of a function f is given below. Sketch a graph of f' using this information.



Example 6.

The graph of a function f is given below. Sketch a graph of f' using this information.



We now return to the discussion of continuity (Section 2.6) and investigate the relationship between continuity and differentiability. Specifically, if a function is differentiable at a point, then it is also continuous at that point. We state this fact now as a theorem:

Theorem 1 (Differentiable Implies Continuous).
If a function f is differentiable at a, then f is continuous at a.

Conversely, if f is not continuous at a, then f is not differentiable at a.

More generally, a function f is not differentiable if the following happens:

Theorem 2 (When Is a Function Not Differentiable at a Point?).
A function f is not differentiable at a point a if at least one of the following conditions holds:

1. *f is not continuous at a*
2. *f has a corner at a*
3. *f has a vertical tangent at a*

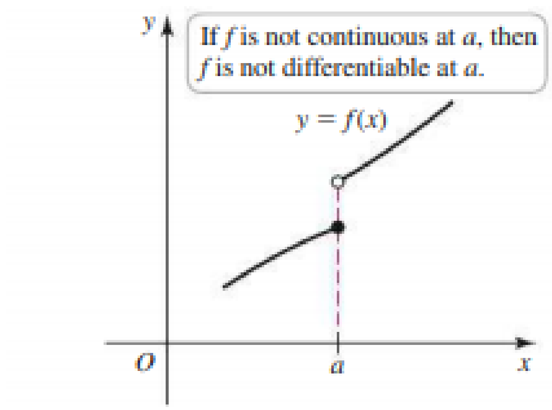


Figure 1: f is not continuous at a

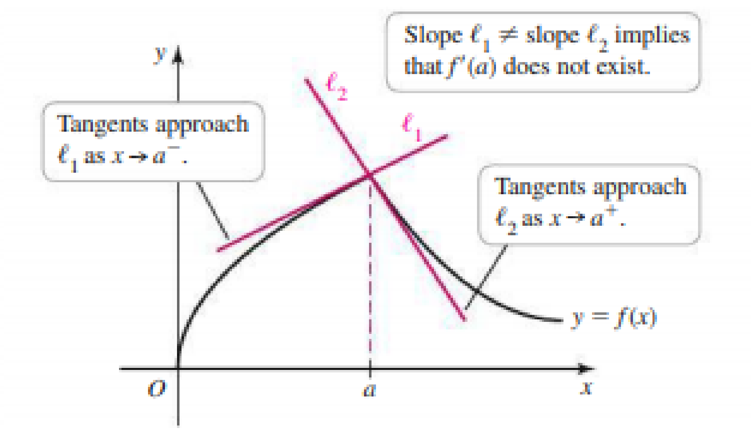


Figure 2: f has a corner at a

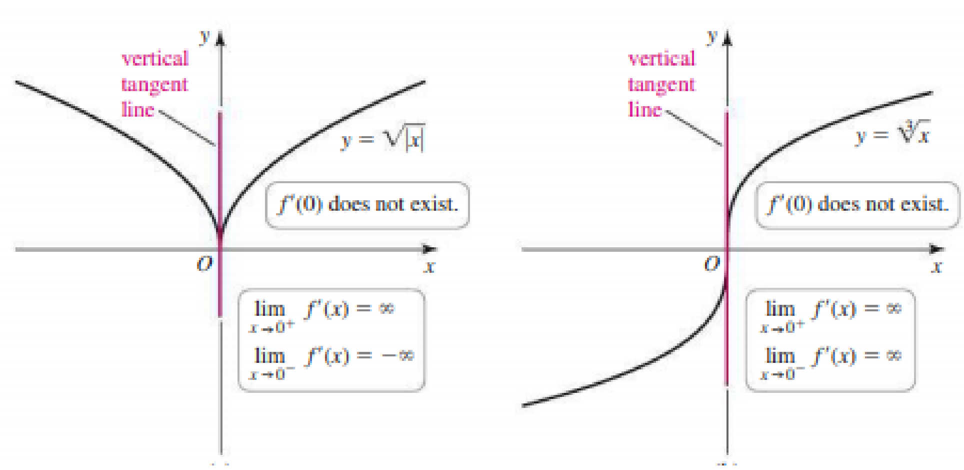


Figure 3: f has a vertical tangent at a

Section 3.3 - Rules of Differentiation

If you always had to use limits to evaluate derivatives, as we did in Section 3.1, calculus would be a tedious affair. The goal of this section is to establish rules and formulas for quickly evaluating derivatives – not just for individual functions but for entire families of functions.

In what follows, we'll begin to use some alternative notation for the derivative. Namely, we will write

$$\frac{d}{dx}\Big(f(x)\Big) = \frac{df}{dx}(x) = f'(x).$$

The term $\frac{d}{dx}$ in the above can be thought of as the "derivative operator." That is, as a *mapping* which sends a given differentiable *function*, f, to its derivative, f'. Later in the course, this notation will become particularly useful to illustrate a variety of concepts.

Theorem 3 (Differentiation Rules).
Let f and g be differentiable at x, let c be a real number and let $n \neq 0$ be a real number. Then, we have the following:

1. $\frac{d}{dx}\Big(c\Big) = 0$ *(Constant Rule)*
2. $\frac{d}{dx}\Big(x^n\Big) = nx^{n-1}$ *(Power Rule)*
3. $\frac{d}{dx}\Big(cf(x)\Big) = c\frac{d}{dx}\Big(f(x)\Big)$ *(Constant Multiple Rule)*
4. $\frac{d}{dx}\Big(f(x)+g(x)\Big) = \frac{d}{dx}\Big(f(x)\Big) + \frac{d}{dx}\Big(g(x)\Big)$ *(Sum Rule)*
5. $\frac{d}{dx}\Big(f(x)-g(x)\Big) = \frac{d}{dx}\Big(f(x)\Big) - \frac{d}{dx}\Big(g(x)\Big)$ *(Difference Rule)*
6. $\frac{d}{dx}\Big(e^x\Big) = e^x$ *(Derivative of e^x)*

Occasionally, we'll also deal with higher-order derivatives, which we now define.

Definition 4 (Higher-Order Derivatives).
Assuming $y = f(x)$ can be differentiated as often as necessary, the second derivative of f is

$$f''(x) = \frac{d}{dx}\Big(f'(x)\Big) = \frac{d}{dx}\left(\frac{d}{dx}\Big(f(x)\Big)\right).$$

For integers $n \geq 2$, the n^{th} derivative of f is

$$f^{(n)}(x) = \frac{d}{dx}\Big(f^{(n-1)}(x)\Big).$$

Example 7.

Find the indicated derivatives:

1. y', where $y = x^3$
2. $\dfrac{dy}{dx}$, where $y = \dfrac{1}{x^7}$
3. $\dfrac{d}{du}\left(2u^2 - 3u + 8\right)$
4. $f'(x)$, where $f(x) = x^{3.2} + 7x^{2.1} - 3x + 7$
5. $\dfrac{d}{dx}\left(\dfrac{1}{x^4} + 6\sqrt{x} - 5x + 8\right)$
6. $f'(x)$, where $f(x) = x^{4/5} + 4x^{3/5} - 2x^{-1/5} + 5$

1. Using the power rule, we find
$$y' = 3x^2$$
2. Re-writing $y = \dfrac{1}{x^7}$ as $y = x^{-7}$, we use the power rule to find
$$\frac{dy}{dx} = -7x^{-8}$$
3. Using the sum rule and power rule, we find
$$\frac{d}{du}\left(2u^2 - 3u + 8\right) = 4u - 3$$
4. Using the sum rule and power rule, we find
$$f'(x) = 3.2x^{2.2} + 14.7x^{1.1} - 3$$
5. Re-writing $\dfrac{1}{x^4} + 6\sqrt{x} - 5x + 8$ as $x^{-4} + 6x^{1/2} - 5x + 8$, we use the sum rule and power rule to find
$$\frac{d}{dx}\left(\frac{1}{x^4} + 6\sqrt{x} - 5x + 8\right) = -4x^{-5} + 3x^{-1/2} - 5$$
6. Using the sum rule and power rule, we find
$$f'(x) = \frac{4}{5}x^{-1/5} + \frac{12}{5}x^{-2/5} + \frac{2}{5}x^{-6/5}$$

Example 8.

Find the equation of the line tangent to the graph of $f(x) = 3e^x - x^2$ at the point $\big(0, f(0)\big)$.

Using Definition 2, we know that the equation of the line tangent to the graph of $f(x)$ at the point $\big(a, f(a)\big)$ is

$$y - f(a) = f'(a)(x - a).$$

Substituting $a = 0$, we then have

$$y - f(0) = f'(0)(x - 0).$$

Evaluating, we find $f(0) = 3e^0 - 0^2 = 3(1) - 0 = 3$. Now, using Theorem 3, we find $f'(x) = 3e^x - 2x$. Evaluating our result, we find $f'(0) = 3e^0 - 2(0) = 3(1) - 0 = 3$. Hence, we may conclude the equation of the line tangent to the graph of $f(x) = 3e^x - x^2$ at the point $\big(0, f(0)\big)$ is

$$y - 3 = 3x, \text{ or, re-writing, } y = 3x + 3.$$

Example 9.

Given the function $f(x) = 7x^2 - 9x + 2$, find:

1. $f'(x)$
2. the slope of the line tangent to the graph of f at the point $(3, f(3))$
3. the value(s) of x for which the line tangent to the graph of f at $(x, f(x))$ is horizontal.

1. Using the sum rule and the power rule, we find $f'(x) = 14x - 9$
2. As the slope of the line tangent to the graph of f at the point $(3, f(3))$ is given by $f'(3)$, we find that $m_{tan} = 14(3) - 9 = 33$
3. As the slope of the line tangent to the graph of f at the point $(x, f(x))$ is given by $f'(x)$, and as a horizontal line has slope zero, we need only solve $f'(x) = 0$. After some routine algebra, we find that the line tangent to the graph of f at $\left(\frac{9}{14}, f\left(\frac{9}{14}\right)\right)$ is horizontal.

Section 3.4 - The Product and Quotient Rules

Theorem 4 (Product Rule).
If f and g are differentiable at x, then

$$\frac{d}{dx}\Big(f(x)g(x)\Big) = f'(x)g(x) + f(x)g'(x).$$

Theorem 5 (Quotient Rule).
If f and g are differentiable at x and $g(x) \neq 0$, then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{\big(g(x)\big)^2}.$$

Example 10.
Find the first derivative of:

1. $f(x) = 4x^5(2x^2 - 13x + 2)$
2. $f(x) = (4x^2 + 7)(2x^2 - 9)$
3. $f(x) = \dfrac{13x}{3x^2 - 2}$

1. Using the product rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(4x^5)(2x^2 - 13x + 2) + (4x^5)\frac{d}{dx}(2x^2 - 13x + 2) \\ &= (20x^4)(2x^2 - 13x + 2) + (4x^5)(4x - 13). \end{aligned}$$

2. Using the product rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(4x^2 + 7)(2x^2 - 9) + (4x^2 + 7)\frac{d}{dx}(2x^2 - 9) \\ &= (8x)(2x^2 - 9) + (4x^2 + 7)(4x). \end{aligned}$$

3. Using the quotient rule,

$$\begin{aligned} f'(x) &= \frac{\frac{d}{dx}(13x)(3x^2 - 2) - (13x)\frac{d}{dx}(3x^2 - 2)}{(3x^2 - 2)^2} \\ &= \frac{13(3x^2 - 2) - (13x)(6x)}{(3x^2 - 2)^2} \\ &= \frac{-39x^2 - 26)}{(3x^2 - 2)^2}. \end{aligned}$$

Example 11.
Find the first derivative of:

1. $f(x) = \dfrac{2x-5}{3x^2+5}$
2. $f(x) = 3e^{2x}$
3. $f(x) = (x+1)^3$

1. Using the quotient rule,

$$\begin{aligned} f'(x) &= \frac{\frac{d}{dx}(2x-5)(3x^2+5)-(2x-5)\frac{d}{dx}(3x^2+5)}{(3x^2+5)^2} \\ &= \frac{2(3x^2+5)-(2x-5)(6x)}{(3x^2+5)^2} \\ &= \frac{-6x^2+30x+10}{(3x^2+5)^2} \end{aligned}$$

2. First, use the properties of exponents to re-write $f(x) = 3e^{2x}$ as $f(x) = 3e^xe^x$. Now, using the product rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3e^x)(e^x) + (3e^x)\frac{d}{dx}(e^x) \\ &= (3e^x)(e^x) + (3e^x)(e^x) \\ &= 6e^x. \end{aligned}$$

3. Using the product rule twice, we find

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x+1)\Big((x+1)^2\Big) + (x+1)\frac{d}{dx}\Big((x+1)^2\Big) \\ &= (x+1)^2 + (x+1)\left(\frac{d}{dx}(x+1)(x+1) + (x+1)\frac{d}{dx}(x+1)\right) \\ &= (x+1)^2 + (x+1)\big(2(x+1)\big) \\ &= 3(x+1)^2. \end{aligned}$$

5

Section 3.5 - Derivatives of Trigonometric Functions

From variations in market trends and ocean temperatures to daily fluctuations in tides and hormone levels, change is often cyclical or periodic. Trigonometric functions are well suited for describing such cyclical behavior. In this section, we investigate the derivatives of trigonometric functions and their many uses.

While we will not specifically show how one is able to obtain the derivatives of $sin(x)$ and $cos(x)$ (which, via the differentiation rules of the previous section, yield the derivatives of $tan(x)$, $sec(x)$, $csc(x)$, and so on), we will touch on two limits which are involved in doing so.

Theorem 6 (Trigonometric Limits).

$$\lim_{x\to 0} \frac{sin(x)}{x} = 1, \qquad \lim_{x\to 0} \frac{cos(x)-1}{x} = 0$$

Following this, let us observe that the trigonometric functions have derivatives as follows:

Theorem 7 (Derivatives of Trigonometric Functions).

1. $\frac{d}{dx}\big(sin(x)\big) = cos(x)$
2. $\frac{d}{dx}\big(cos(x)\big) = -sin(x)$
3. $\frac{d}{dx}\big(tan(x)\big) = sec^2(x)$
4. $\frac{d}{dx}\big(sec(x)\big) = sec(x)tan(x)$
5. $\frac{d}{dx}\big(cot(x)\big) = -csc^2(x)$
6. $\frac{d}{dx}\big(csc(x)\big) = -csc(x)cot(x)$

Example 12.

Evaluate the limit: $\lim\limits_{x\to 0} \frac{sin(4x)}{x}$.

First, we wish to evaluate $\lim\limits_{x\to 0} \frac{sin(4x)}{x}$. To use the fact that $\lim\limits_{x\to 0} \frac{sin(x)}{x} = 1$, the argument of the sine function in the numerator must be the same as the denominator. Multiplying and dividing $\frac{sin(4x)}{x}$ by 4, we evaluate the limit as follows:

$$\begin{aligned} \lim_{x\to 0} \frac{sin(4x)}{x} &= \lim_{x\to 0} \frac{4sin(4x)}{4x} \\ &= \lim_{x\to 0} 4\frac{sin(4x)}{4x} \\ &= 4\lim_{x\to 0} \frac{sin(4x)}{4x} \\ &= 4\lim_{t\to 0} \frac{sin(t)}{t} \qquad \text{(setting } t = 4x\text{)} \\ &= 4(1) = 4. \end{aligned}$$

Example 13.

Evaluate the limit $\lim_{x\to 0} \frac{sin(3x)}{sin(5x)}$.

We wish to evaluate $\lim_{x\to 0} \frac{sin(3x)}{sin(5x)}$. We will first divide both the numerator and denominator by x, then proceed to evaluate the limit as follows:

$$\begin{aligned}
\lim_{x\to 0} \frac{sin(3x)}{sin(5x)} &= \lim_{x\to 0} \frac{\frac{sin(3x)}{x}}{\frac{sin(5x)}{x}} \\
&= \frac{\lim_{x\to 0} \frac{3sin(3x)}{3x}}{\lim_{x\to 0} \frac{5sin(5x)}{5x}} \\
&= \frac{3\lim_{x\to 0} \frac{sin(3x)}{3x}}{5\lim_{x\to 0} \frac{sin(5x)}{5x}} \\
&= \frac{3\lim_{t\to 0} \frac{sin(t)}{t}}{5\lim_{w\to 0} \frac{sin(w)}{w}} \qquad \text{(setting } t = 3x \text{ and } w = 5x\text{)} \\
&= \frac{3(1)}{5(1)} = \frac{3}{5}.
\end{aligned}$$

Example 14.

Find the derivative of $y = sec(x)csc(x)$.

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}\Big(sec(x)csc(x)\Big) \\
&= \frac{d}{dx}\Big(sec(x)\Big)\Big(csc(x)\Big) + \Big(sec(x)\Big)\frac{d}{dx}\Big(csc(x)\Big) \\
&= \Big(sec(x)tan(x)\Big)\Big(csc(x)\Big) + \Big(sec(x)\Big)\Big(-csc(x)cot(x)\Big) \\
&= \frac{1}{cos(x)}\frac{sin(x)}{cos(x)}\frac{1}{sin(x)} - \frac{1}{cos(x)}\frac{1}{sin(x)}\frac{cos(x)}{sin(x)} \\
&= \frac{1}{cos^2(x)} - \frac{1}{sin^2(x)} \\
&= sec^2(x) - csc^2(x).
\end{aligned}$$

Section 3.6 - Derivatives as Rates of Change

The theme of this section is the *derivative* as a *rate of change*. Observing the world around us, we see that almost everything is in a state of change: The size of the Internet is increasing; your blood pressure fluctuates; as supply increases, prices (usually) decrease; and the universe is expanding. This section explores a few of the many applications of this idea and demonstrates why calculus is sometimes referred to as the mathematics of change.

One such example we have seen before: Position, speed, velocity, and acceleration.

Definition 5 (Velocity, Speed, and Acceleration).
Suppose an object moves along a line with position $s = f(t)$. Then:

$$\begin{aligned} \text{the velocity at time } t \text{ is} \quad & v = \frac{ds}{dt} = f'(t). \\ \text{the speed at time } t \text{ is} \quad & |v| = |f'(t)|. \\ \text{the acceleration at time } t \text{ is} \quad & a = \frac{dv}{dt} = \frac{d^2s}{dt^2} = f''(t). \end{aligned}$$

Another common application occurs in economics. Particularly, we can see the derivative highlighted in the related notions of cost functions, marginal cost functions, and average costs functions.

Definition 6 (Average and Marginal Cost).
The cost function $C(x)$ gives the cost to produce the first x items in a manufacturing process. The average cost to produce x items is $\overline{C}(x) = C(x)/x$. The marginal cost $C'(x)$ is the approximate cost to produce one additional item after producing x items.

Similarly, we see the derivative appear in the economic notion of elasticity.

Definition 7 (Elasticity).
If the demand for a product varies with the price according the function $D = f(p)$, then the price elasticity of the demand is

$$E(p) = \frac{dD}{dp}\frac{p}{D} = \frac{pf'(p)}{f(p)}.$$

When $E < -1$, the demand is said to be elastic. When $-1 < E < 0$, the demand is said to be inelastic. When $E = -1$, the demand is said to be unit elastic.

Example 15. Suppose a stone is thrown vertically upward with an initial velocity of 64 ft¿s from a bridge 96 ft above a river. By Newton's laws of motion, the position of the stone (measured as the height above the river) after t seconds is

$$s(t) = -16t^2 + 64t + 96,$$

where $s = 0$ is the level of the river.

1. Find the velocity and acceleration functions.
2. What is the highest point above the river reached by the stone?
3. With what velocity will the stone strike the river?

1. By Definition 5, $v = -32t + 64$ and $a = -32$.
2. When the stone reaches its high point, its velocity is zero. Solving

   $$v(t) = -32t + 64 = 0$$

   yields $t = 2$; therefore, the stone reaches its maximum height 2 seconds after it is thrown. Its height (in feet) at that instant is

   $$s(2) = -16(2)^2 + 64(2) + 96 = 160.$$

3. To determine the velocity at which the stone strikes the river, we first determine when it reaches the river. The stone strikes the river when

   $$s(t) = -16t^2 + 64t + 96 = 0.$$

   Dividing both sides of the equation by -16, we obtain

   $$t^2 - 4t - 6 = 0.$$

   Using the quadratic formula, the solutions are $t \approx 5.16$ or $t \approx -1.16$. Because the stone is thrown at $t = 0$, only positive values of t are of interest; therefore, the relevant root is $t \approx 5.16$. The velocity of the stone (in ft/s) when it strikes the river is approximately

   $$v(5.16) = -32(5.16) + 64 = -101.12.$$

Example 16. Suppose the cost of producing x items is given by the function

$$C(x) = -0.02x^2 + 50x + 100, \text{ for } 0 \le x \le 1000.$$

1. Determine the average and marginal cost functions.
2. Determine the average and marginal cost for the first 100 items and interpret these values.

1. The average cost is
$$\overline{C}(x) = \frac{C(x)}{x} = \frac{-0.02x^2 + 50x + 100}{x}.$$
The marginal cost is
$$\frac{dC}{dx} = -0.04x + 50.$$
So, the average cost decreases as the number of items produced increases, and the marginal cost decreases linearly with a slope of -0.04.
2. To produce $x = 100$ items, the average cost is
$$\overline{C}(100) = \frac{C(100)}{100} = \frac{-0.02(100)^2 + 50(100) + 100}{100} = \$49/\text{item}.$$
and the marginal cost is
$$C'(100) = -0.04(100) + 50 = \$46/\text{item}.$$
These results mean that the average cost of producing the first 100 items is \$ 49 per item, but the cost of producing one additional item (the 101^{st} item) is approximately \$46. Therefore, producing one more item is less expensive than the average cost of producing the first 100 items.

Example 17. The demand for processed pork in Canada is described by the function $D(p) = 286 - 20p$. Find the price elasticity of demand and determine the price levels at which the demand for pork is elastic and the price levels at which the demand for pork is inelastic.

The price elasticity of demand for pork is

$$E(p) = \frac{pD'(p)}{D(p)} = \frac{p(-20)}{286 - 20p} = \frac{-10p}{143 - 10p}.$$

As demand is elastic when $E(p) < -1$ and inelastic when $-1 < E(p) < 0$, we correspondingly solve $\dfrac{-10p}{143 - 10p} = -1$. Solving, $p = \dfrac{143}{20} = \$7.15$. So, after constructing a sign chart, we may conclude that demand is elastic when $p > \$7.15$ and inelastic when $\$0 < p < \$7.15$

7

Section 3.7 - The Chain Rule

The differentiation rules presented so far allow us to find derivatives of many functions. However, these rules are inadequate for finding the derivatives of most composite functions. In this chapter, we will describe efficient method for differentiating composite functions called the Chain Rule.

Theorem 8 (The Chain Rule).
Let $y = f(u)$, where $u = g(x)$. Suppose g is differentiable at x and suppose f is differentiable at $u = g(x)$. Then,

$$\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big)g'(x).$$

Or, stating the above in another way (primarily to highlight the usefulness of Liebniz notation), we have

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

Example 18. Find $\frac{dy}{dx}$, where 1. $y = (5x+4)^3$ 2. $y = sin^3(x)$ 3. $y = sin\big(x^3\big)$.

1. First, we find a way to write the function $y = (5x+4)^3$ as a composition. Setting $y(u) = u^3$ and $u = 5x+4$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 3u^2$ and $\frac{du}{dx} = 5$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx}\\ &= 15u^2 = 15(5x+4)^2.\end{aligned}$$
2. First, we find a way to write the function $y = sin^3(x)$ as a composition. Setting $y(u) = u^3$ and $u = sin(x)$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 3u^2$ and $\frac{du}{dx} = cos(x)$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx}\\ &= 3u^2cos(x) = 3cos(x)sin^2(x).\end{aligned}$$
3. First, we find a way to write the function $y = sin\big(x^3\big)$ as a composition. Setting $y(u) = sin(u)$ and $u = x^3$ will suffice for our purposes. Calculating, $\frac{dy}{du} = cos(u)$ and $\frac{du}{dx} = 3x^2$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx}\\ &= cos(u)3x^2 = 3x^2cos\big(x^3\big).\end{aligned}$$

Example 19. Find $\frac{dy}{dx}$, where 1. $y = (6x^3 + 3x + 1)^{10}$ 2. $y = \sqrt{5x^2 + 1}$ 3. $y = \left(\frac{5t^2}{3t^2 + 1}\right)^3$.

1. First, we find a way to write the function $y = (6x^3 + 3x + 1)^{10}$ as a composition. Setting $y(u) = u^{10}$ and $u = 6x^3 + 3x + 1$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 10u^9$ and $\frac{du}{dx} = 18x^2 + 3$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} \\ &= 10u^9(18x^2 + 3) \\ &= 10(6x^3 + 3x + 1)^9(18x^2 + 3).\end{aligned}$$

2. First, we find a way to write the function $y = \sqrt{5x^2 + 1}$ as a composition. Setting $y(u) = \sqrt{u}$ and $u = 5x^2 + 1$ will suffice for our purposes. Calculating, $\frac{dy}{du} = \frac{1}{2\sqrt{u}}$ and $\frac{du}{dx} = 10x$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} \\ &= \frac{1}{2\sqrt{u}}(10x) \\ &= \frac{5x}{\sqrt{5x^2 + 1}}.\end{aligned}$$

3. First, we find a way to write the function $y = \left(\frac{5t^2}{3t^2 + 1}\right)^3$ as a composition. Setting $y(u) = u^3$ and $u = \frac{5t^2}{3t^2 + 1}$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 3u^2$ and $\frac{du}{dx} = \frac{10t(3t^2 + 1) - (5t^2)(6t)}{(3t^2 + 1)^2}$. Hence, we then have that
$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} \\ &= 3u^2\frac{10t(3t^2 + 1) - (5t^2)(6t)}{(3t^2 + 1)^2} \\ &= 3\left(\frac{5t^2}{3t^2 + 1}\right)^2\frac{10t(3t^2 + 1) - (5t^2)(6t)}{(3t^2 + 1)^2} \\ &= \frac{1500t^5}{(3t^2 + 2)^4}.\end{aligned}$$

Example 20. Find $\frac{d}{dx}\left[\left(tan(x)+10\right)^{21}\right]$.

First, we find a way to write the function $y = \left(tan(x)+10\right)^{21}$ as a composition. Setting $y(u) = u^{21}$ and $u = tan(x)+10$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 21u^{20}$ and $\frac{du}{dx} = sec^2(x)$. Hence, we then have that

$$\begin{aligned}\frac{d}{dx}\left[\left(tan(x)+10\right)^{21}\right] &= \frac{dy}{dx}\\ &= \frac{dy}{du}\frac{du}{dx}\\ &= 21u^{20}sec^2(x)\\ &= 21\left(tan(x)+10\right)^{20}sec^2(x).\end{aligned}$$

Example 21. Find $\frac{d}{dx}\left[sin\left(e^{cos(x)}\right)\right]$.

In this problem, we will have to make use of the chain rule twice. Let's begin as normal. First, we find a way to write the function $y = sin\left(e^{cos(x)}\right)$ as a composition. Setting $y(u) = sin(u)$ and $u = e^{cos(x)}$ will suffice for our purposes. Calculating, $\frac{dy}{du} = cos(u)$. We'll need to find $\frac{du}{dx}$ in a moment, but for now, let's observe that we have

$$\begin{aligned}\frac{d}{dx}\left[sin\left(e^{cos(x)}\right)\right] = \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx}\\ &= cos(u)\frac{du}{dx} = cos\left(e^{cos(x)}\right)\frac{du}{dx}.\end{aligned}$$

so, we can finish this problem in short order if we simply find $\frac{du}{dx}$. To do so, we'll find a way to write the function $u = e^{cos(x)}$ as a composition. Setting $u(w) = e^w$ and $w = cos(x)$ will suffice for our purposes. Calculating, $\frac{du}{dw} = e^w$ and $\frac{dw}{dx} = -sin(x)$. So,

$$\begin{aligned}\frac{du}{dx} &= \frac{du}{dw}\frac{dw}{dx}\\ &= e^w(-sin(x))\\ &= -sin(x)e^{cos(x)}.\end{aligned}$$

So, we may now see that

$$\frac{d}{dx}\left[sin\left(e^{cos(x)}\right)\right] = -sin(x)cos\left(e^{cos(x)}\right)sin(x)e^{cos(x)}.$$

Section 3.8 - Implicit Differentiation

This chapter has been devoted to calculating derivatives of functions of the form $y = f(x)$, where y is defined *explicitly* as a function of x. However, relations between variables are often expressed *implicitly*. For example, the equation of the unit circle $x^2 + y^2 = 1$ does not specify y *directly*, but rather defines y *implicitly*.

If we start with such an expression and want to find y', one might go about this by solving the equation for y and differentiating. However, as the unit circle case $x^2 + y^2 = 1$ demonstrates, there may be multiple possible solutions for y and, correspondingly, multiple possible derivatives (go ahead, try it out). To avoid annoyances and ambiguities like this, we'll need to develop a new technique. That is, the goal of this section is to find a *single expression* for the derivative *directly* from an equation without first solving for y. This technique is called *implicit differentiation* and, as we will see, proves incredibly useful in a wide variety of applications.

Example 22. Calculate $\dfrac{dy}{dx}$ given that y satisfies the equation $x^2 + y^2 = 1$. Then, find $\dfrac{dy}{dx}$ at the point $\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right)$.

To find $\dfrac{dy}{dx}$, we'll differentiate both sides of the equation. Remember, if f is a function of x, then the chain rule tells us that $\dfrac{d}{dx}\left[\big(f(x)\big)^n\right] = nf'(x)\big(f(x)\big)^{n-1}$. As y denotes a function of x, we'll need to use this. Let's see what happens:

$$\begin{aligned}
\frac{d}{dx}\left(x^2 + y^2\right) &= \frac{d}{dx}(1) \\
2x + \frac{d}{dx}(y^2) &= 0 \quad \text{(differentiating the terms which do not involve } y\text{)} \\
2x + 2y\frac{dy}{dx} &= 0 \quad \text{(applying the chain rule, as } y \text{ is a function of } x\text{)} \\
\frac{dy}{dx} &= \frac{-2x}{2y} \\
\frac{dy}{dx} &= \frac{-x}{y}.
\end{aligned}$$

So, at the point $\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right)$, we have that

$$\frac{dy}{dx} = \frac{-\left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)} = \frac{-1}{\sqrt{3}}.$$

Example 23. Find y' given that y satisfies the equation $sin(xy) = x^2 + y$.

$$\begin{aligned}
\frac{d}{dx}(sin(xy)) &= \frac{d}{dx}\left(x^2+y\right)\\
cos(xy)(y + x\frac{dy}{dx}) &= 2x + \frac{dy}{dx}\\
ycos(xy) + xcos(xy)\frac{dy}{dx} &= 2x + \frac{dy}{dx}\\
xcos(xy)\frac{dy}{dx} - \frac{dy}{dx} &= 2x - ycos(xy)\\
\frac{dy}{dx}(xcos(xy) - 1) &= 2x - ycos(xy)\\
\frac{dy}{dx} &= \frac{2x - ycos(xy)}{xcos(xy) - 1}
\end{aligned}$$

So, $y' = \dfrac{2x - ycos(xy)}{xcos(xy) - 1}$.

Example 24. Find $\dfrac{dy}{dx}$ at $(2,-1)$ given that $(x^2+y^2)^2 = \dfrac{25}{3}(x^2-y^2)$.

$$\begin{aligned}
\frac{d}{dx}\left[(x^2+y^2)^2\right] &= \frac{d}{dx}\left[\frac{25}{3}(x^2-y^2)\right]\\
2(x^2+y^2)\frac{d}{dx}\left[(x^2+y^2)\right] &= \frac{25}{3}\left(\frac{d}{dx}(x^2) - \frac{d}{dx}(y^2)\right)\\
2(x^2+y^2)\left(2x + 2y\frac{dy}{dx}\right) &= \frac{25}{3}\left(2x - 2y\frac{dy}{dx}\right)\\
4x(x^2+y^2) + 4y(x^2+y^2)\frac{dy}{dx} &= \frac{50}{3}x - \frac{50}{3}y\frac{dy}{dx}\\
4y(x^2+y^2)\frac{dy}{dx} + \frac{50}{3}y\frac{dy}{dx} &= \frac{50}{3}x - 4x(x^2+y^2)\\
\frac{dy}{dx}\left(4y(x^2+y^2) + \frac{50}{3}y\right) &= \frac{50}{3}x - 4x(x^2+y^2)\\
\frac{dy}{dx} &= \frac{\frac{50}{3}x - 4x(x^2+y^2)}{4y(x^2+y^2) + \frac{50}{3}y}
\end{aligned}$$

So, at $(2,-1)$, plugging in $x = 2$ and $y = -1$ and calculating yields that $\dfrac{dy}{dx} = \dfrac{2}{11}$.

Section 3.9 - Derivatives of Logarithmic and Exponential Functions

We return now to the major theme of this chapter: developing rules of differentiation for the standard families of functions. First, we discover how to differentiate the natural logarithmic function. From there, we treat general exponential and logarithmic functions.

Theorem 9 (Derivative of $ln(x)$).

1. $\frac{d}{dx}(ln(x)) = \frac{1}{x}$ *for* $x > 0$
2. $\frac{d}{dx}\left(ln\left(|x|\right)\right) = \frac{1}{x}$ *for* $x \neq 0$
3. $\frac{d}{dx}\left(ln\left(|u(x)|\right)\right) = \frac{u'(x)}{u(x)}$ *if* u *is differentiable at* x *and* $u(x) \neq 0$.

Example 25. Find $\frac{dy}{dx}$ for the following functions:

$$1.\ y = ln(4x) \quad 2.\ y = xln(x) \quad 3.\ y = ln\left(|sec(x)|\right) \quad 4.\ y = \frac{ln(x^2)}{x^2}$$

1. Using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[ln(4x)] \\ &= \frac{4}{4x} \\ &= \frac{1}{x}\end{aligned}$$

2. Using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[xln(x)] \\ &= \left(ln(x) + x\frac{1}{x}\right) \\ &= ln(x) + 1\end{aligned}$$

3. Using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left[ln\left(|sec(x)|\right)\right] \\ &= \frac{1}{sec(x)}\frac{d}{dx}\left(sec(x)\right) \\ &= \frac{1}{sec(x)}sec(x)tan(x) \\ &= tan(x)\end{aligned}$$

4. Using the quotient rule and the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left[\frac{ln(x^2)}{x^2}\right] \\ &= \frac{x^2\left(\frac{1}{x^2}2x\right) - ln\left(x^2\right)2x}{\left(x^2\right)^2} \\ &= \frac{2x - 2xln\left(x^2\right)}{x^4} \\ &= \frac{2 - 2ln\left(x^2\right)}{x^3}\end{aligned}$$

Theorem 10 (Derivative of b^x).
If $b > 0$ and $b \neq 1$, then for all x,

$$\frac{d}{dx}\left(b^x\right) = b^x ln(b).$$

Example 26. Find the derivative of the following functions:

$$1.\ \ f(x) = 3^x \qquad 2.\ \ g(t) = 108\left(2^{t/12}\right)$$

1. By Theorem 10, $f'(x) = 3^x ln(3)$.
2. By the chain rule and Theorem 10,

$$g'(t) = 108\frac{d}{dt}\left(2^{t/12}\right) = 108\ \ ln\left[2\left(2^{t/12}\right)\right]\frac{1}{12} = 9\ \ ln\left[2\left(2^{t/12}\right)\right].$$

Example 27. Let $f(x) = x^{sin(x)}$. Find $f'(x)$.

First, let us note that $b^x = e^{xln(b)}$. So, using this, we may rewrite f as

$$f(x) = x^{sin(x)} = e^{sin(x)ln(x)}.$$

So, by the chain rule,

$$\begin{aligned}
f'(x) &= \frac{d}{dx}\left[e^{sin(x)ln(x)}\right] \\
&= e^{sin(x)ln(x)}\frac{d}{dx}\left[sin(x)ln(x)\right] \\
&= e^{sin(x)ln(x)}\left(\frac{sin(x)}{x} + cos(x)ln(x)\right) \\
&= x^{sin(x)}\left(\frac{sin(x)}{x} + cos(x)ln(x)\right).
\end{aligned}$$

Section 3.10 - Derivatives of Inverse Trigonometric Functions

The inverse trigonometric functions are major players in calculus. In this section, we develop the derivatives of the six inverse trigonometric functions and begin an exploration of their many applications. The method for differentiating the inverses of more general functions is also presented.

Theorem 11 (Derivatives of Inverse Trigonometric Functions).

$$\frac{d}{dx}\left[sin^{-1}(x)\right] = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\left[cos^{-1}(x)\right] = \frac{-1}{\sqrt{1-x^2}},\ for\ \ -1 < x < 1$$
$$\frac{d}{dx}\left[tan^{-1}(x)\right] = \frac{1}{1+x^2} \qquad \frac{d}{dx}\left[cot^{-1}(x)\right] = \frac{-1}{1+x^2},\ for\ \ -\infty < x < \infty$$
$$\frac{d}{dx}\left[sec^{-1}(x)\right] = \frac{1}{|x|\sqrt{x^2-1}} \qquad \frac{d}{dx}\left[csc^{-1}(x)\right] = \frac{-1}{|x|\sqrt{x^2-1}},\ for\ \ |x| > 1$$

Theorem 12 (Derivative of Inverse Function).
Let f be differentiable and have an inverse on an interval I. *If x_0 is a point of* I *at which $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and*

$$\left(f^{-1}\right)'(y_0) = \frac{1}{f'(x_0)},\ \ where\ \ y_0 = f(x_0).$$

Example 28.
Evaluate $f'(2\sqrt{3})$, where $f(x) = x\ \ tan^{-1}(x/2)$.

Using Theorem 11, we proceed to calculate $f'(x)$ as follows:

$$f'(x) = tan^{-1}(x/2) + x\frac{1}{1+(x/2)^2}\frac{1}{2} \quad \text{(Product rule and chain rule)}$$
$$= tan^{-1}\left(\frac{x}{2}\right) + \frac{2x}{4+x^2}$$

So, recalling that $tan^{-1}(\sqrt{3}) = \pi/3$, we may see that

$$f'(2\sqrt{3}) = tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) + \frac{2(2\sqrt{3})}{4+(2\sqrt{3})^2} = \frac{\pi}{3} + \frac{\sqrt{3}}{4}$$

Example 29.
Find the equation of the line tangent to the graph of $g(x) = sec^{-1}(2x)$ at the point $(1, \pi/3)$.

The slope of the tangent line at $(1, \pi/3)$ is $g'(1)$. Using the Chain Rule, we have

$$\begin{aligned} g'(x) &= \frac{d}{dx}\left(sec^{-1}(2x)\right) \\ &= \frac{2}{|2x|\sqrt{4x^2-1}} \\ &= \frac{1}{|x|\sqrt{4x^2-1}}. \end{aligned}$$

So, $g'(1) = 1/\sqrt{3}$, and the equation of the tangent line is

$$y - \frac{\pi}{3} = \frac{1}{\sqrt{3}}\left(x-1\right).$$

Example 30.
The function $f(x) = \sqrt{x} + x^2 + 1$ is one-to-one for $x \geq 0$ and has an inverse on that interval. Find the slope of the curve $y = f^{-1}(x)$ at the point $(3, 1)$.

The point $(1, 3)$ is on the graph of f; therefore, $(3, 1)$ is on the graph of f^{-1}. In this case, the slope of the curve $y = f^{-1}(x)$ at the point $(3, 1)$ is the reciprocal of the slope of the curve $y = f(x)$ at $(1, 3)$. Note that

$$f'(x) = \frac{1}{2\sqrt{x}} + 2x$$

which means that $f'(1) = \frac{1}{2} + 2 = \frac{5}{2}$. Therefore,

$$\left(f^{-1}\right)'(3) = \frac{1}{f'(1)} = \frac{1}{5/2} = \frac{2}{5}.$$

Observe that it is not necessary to find a formula for f^{-1} to evaluate its derivative at a point.

11

Section 3.11 - Related Rates

We now return to the theme of derivatives as rates of change in problems in which the variables change with respect to time. The essential feature of these problems is that two or more variables, which are related in a known way, are themselves changing in time. Let's look at an example.

Example 31.
An oil rig springs a leak in calm seas, and the oil spreads in a circular patch around the rig. If the radius of the oil patch increases at a rate of $30 m/hr$, how fast is the area of the patch increasing when the patch has a radius of 100 meters?

Two variables change simultaneously: the radius of the circle and its area. The key relationship between the radius and area is $A = \pi r^2$. It helps to rewrite the basic relationship showing explicitly which quantities vary in time. In this case, we rewrite A and r as $A(t)$ and $r(t)$ to emphasize that they change with respect to t (time). The general expression relating the radius and area at any time t is $A(t) = \pi\left(r(t)\right)^2$.

So, let's say that the observed values occur at some fixed time T. The goal is to find the rate of change of the area of the circle at time T, which is $A'(T)$, given that $r'(T) = 30\ \ m/hr$ when $r(T) = 100\ \ m$. That is, we're assuming we know the value of r and r' at some time T, and we want to use this to find the area at time T. To introduce derivatives into the problem, we differentiate the area relation $A(t) = \pi\left(r(t)\right)^2$ with respect to t:

$$\begin{aligned} A'(t) &= \frac{d}{dt}\left[\pi\left(r(t)\right)^2\right] \\ &= \pi\frac{d}{dt}\left[\left(r(t)\right)^2\right] \\ &= \pi 2r(t)r'(t) \\ &= 2\pi r(t)r'(t). \end{aligned}$$

So, using this relationship that holds at all times t, we can substitute the given values of $r(T) = 100\ \ m$ and $r'(T) = 30\ \ m/hr$ and find the rate of change of the area at the specific time T, i.e. at the time the other observations were made:

$$\begin{aligned} A'(T) &= 2\pi r(T)r'(T)\ \ m^2/hr \\ &= 2\pi(100)(30)\ \ m^2/hr \\ &= 6000\pi\ \ m^2/hr. \end{aligned}$$

Notice: what the value of T is, itself, doesn't really matter – we just want to distinguish that the radius of the oil patch was $100\ \ m$ *at some time* instead of *at every time*. This makes sense, since any constant radius function $r(t) = c$ would necessarily have a derivative $r'(t)$ of zero (which would contradict our observed value).

Theorem 13 (Procedure for Solving Related Rates Problems).

1. *Read the problem carefully, making a sketch to organize the given information. Identify the rates that are given and the rate that is to be determined.*
2. *Write one or more equations that express the basic relationships among the variables.*
3. *Introduce rates of change by differentiating the appropriate equation(s) with respect to time t.*
4. *Substitute known values and solve for the desired quantity.*
5. *Check that units are consistent and the answer is reasonable. (For example, does it have the correct sign?)*

Example 32.
Two small planes approach an airport, one flying due west at 120 mi/hr and the other flying due north at 150 mi/hr. Assuming they fly at the same constant elevation, how fast is the distance between the planes changing when the westbound plane is 180 miles from the airport and the northbound plane is 225 miles from the airport?

Let $x(t)$ and $y(t)$ denote the distance from the airport to the westbound and northbound planes, respectively. The paths of the two planes form the legs of a right triangle, and the distance between them, which we'll denote $z(t)$, is the hypotenuse. By the Pythagorean theorem, we then know that $z^2 = x^2 + y^2$.

Our aim is to find dz/dt, the rate of change of the distance between the planes. We first differentiate both sides of $z^2 = x^2 + y^2$ with respect to t:

$$\begin{aligned}
\frac{d}{dt}\left[z^2\right] &= \frac{d}{dt}\left[x^2 + y^2\right] \\
2z\frac{dz}{dt} &= 2x\frac{dx}{dt} + 2y\frac{dy}{dt} \\
\frac{dz}{dt} &= \frac{2x\frac{dx}{dt} + 2y\frac{dy}{dt}}{2z} \\
\frac{dz}{dt} &= \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{z}.
\end{aligned}$$

Notice that the Chain Rule is needed because x, y, and z are functions of t. This equation relates the unknown rate dz/dt to the known quantities x, y, z, dx/dt, and dy/dt. For the westbound plane, $dx/dt = -120\ mi/hr$ (negative because the distance is decreasing), and for the northbound plane, $dy/dt = -150\ mi/hr$. At the moment of interest, when $x = 180\ mi$ and $y = 225\ mi$, the distance between the planes is

$$z = \sqrt{x^2 + y^2}\ mi = \sqrt{180^2 + 225^2}\ mi \approx 288\ mi.$$

So, substituting, we then have that

$$\frac{dz}{dt} \approx \frac{(180\ mi)(-120\ mi/hr) + (225\ mi)(-150\ mi/hr)}{288\ mi} = -192\ mi/hr.$$

Notice that $dz/dt < 0$, which means the distance between the planes is *decreasing* at a rate of about 192 mi/hr.

Example 33.

Sand falls from an overhead bin, accumulating in a conical pile with a radius that is always three times its height. If the sand falls from the bin at a rate of 120 ft^3/min, how fast is the height of the sandpile changing when the pile is 10 ft high?

The three relevant variables are: the volume V, the radius r, and the height h of the sandpile. The aim is to find the rate of change of the height dh/dt at the instant that $h = 10$ ft, given that $dV/dt = 120$ ft^3/min. The basic relationship among the variables is the formula for the volume of a cone, $V = (1/3)\pi r^2 h$. We now use the given fact that the radius is always three times the height. Substituting $r = 3h$ into the volume relationship gives V in terms of h:

$$V = \frac{1}{3}\pi(3h)^2 h = 3\pi h^3.$$

Rates of change are introduced by differentiating both sides of $V = 3\pi h^3$ with respect to t. Using the Chain Rule, we have:

$$\frac{dV}{dt} = 9\pi h^2 \frac{dh}{dt}.$$

Now we find dh/dt at the instant that $h = 10$ ft, given that $dV/dt = 120$ ft^3/min. Solving for dh/dt, we have

$$\frac{dh}{dt} = \frac{\frac{dV}{dt}}{9\pi h^2}.$$

Substituting our values, we have

$$\frac{dh}{dt} = \frac{120\ ft^3/min}{9\pi(10\ ft)^2} \approx 0.042\ ft/min.$$

So, at the instant that the sandpile is 10 ft high, the height is changing at a rate of 0.042 ft min. Notice how the units work out consistently.